

5	(a)	force per unit positive charge acting on a stationary charge	B1	[1]
	(b)	(i)		
		$E = Q / 4\pi\epsilon_0 r^2$	C1	
		$Q = 1.8 \times 10^4 \times 10^2 \times 4\pi \times 8.85 \times 10^{-12} \times (25 \times 10^{-2})^2$	M1	
		$Q = 1.25 \times 10^{-5} \text{ C} = 12.5 \mu\text{C}$	A0	[2]
		(ii)		
		$V = Q / 4\pi\epsilon_0 r$		
		$= (1.25 \times 10^{-5}) / (4\pi \times 8.85 \times 10^{-12} \times 25 \times 10^{-2})$	C1	
		$= 4.5 \times 10^5 \text{ V}$	A1	[2]
		<i>(Do not allow use of $V = Er$ unless explained)</i>		

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6 (a) (i) peak voltage = 4.0 V A1 [1]